



LR2021 LoRa® Performance

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Introduction

LoRa® is a long-range communication technology developed by Semtech. It is a spread spectrum modulation technique derived from chirp spread spectrum (CSS) technology. In 2013, Semtech launched the first generation of LoRa chips, SX1272/76. The second generation, SX1261/62/SX1280, and third generation, LR1110/1120/1121, further enhanced performance.

The LR2021 is the first chip in Semtech's LoRa Plus™ family, incorporating Generation 4 LoRa IP technology. It provides single bill of materials matching for all output power values, an improved power amplifier efficiency, and a flexible packet engine, supporting numerous modulations schemes. This allows PHY-layer compatibility with multiple protocols in addition to LoRaWAN®, such as Sigfox, WM-Bus, Wi-SUN®, Bluetooth® Low Energy.

The LR2021 also maintains backward compatibility with existing LoRaWAN® deployments while introducing revolutionary improvements in sensitivity, data rates, and interference immunity, which are described in this document.

LoRa Overview

There are several key parameters that determine the characteristics of a LoRa signal. These parameters include:

- **Spreading factor (SF):** The spreading factor controls the chirp rate and therefore controls the speed of data transmission. Lower spreading factors mean faster chirps and therefore a higher data transmission rate, while higher spreading factors mean better receive sensitivity, and therefore longer communication distance. LoRa devices support SF5 to SF12, which are also used for LoRaWAN since Regional Parameters RP002-1.0.5.
- **Signal Bandwidth (BW):** The bandwidth corresponds to the frequency variation of the LoRa chirps, and therefore the spectrum bandwidth in kHz. Only a subset of bandwidths are usable in the context of LoRaWAN (125, 250, or 500 kHz), however, the sub-GHz LoRa devices support bandwidths as low as 7.81 kHz, or up to 1000 kHz. LoRa Generation 2 added 2.4GHz ISM band operation capability to LoRa through SX1280, for which certain bandwidths were specific to 2.4 GHz, such as 203, 406, or 812 kHz. LoRa Generations 3 and 4 allow selection for all the available bandwidths independent of the RF frequency configured.
- **Coding rate (CR):** LoRa has forwarding error correction, providing further robustness to errors caused by the environment. The coding rate parameter is configurable from rate 4/5 to rate 1/2.

Benefits

Long Range

LoRa enables kilometres of communication distance. The range is a result of the radio frequency (RF) power emitted by the transmitting device and the receiver sensitivity. The link budget is expressed as follows:

$$\text{Link Budget (dB)} = \text{TX power} - \text{Sensitivity}$$

The transmit power (TX power) correlates with the current consumption, and therefore, the battery lifetime. In Europe, the TX power emitted by an application is limited to +14 dBm. In the United States and India, the maximum output power allowed for a LoRa device is increased to +30 dBm conducted.

LoRa detects the incoming signal as low as 22 dB below the noise floor, achieving unique receiver sensitivities, down to -141 dBm (for LR11xx, 868/915MHz, SF12, BW 125 kHz). This allows LoRa devices to achieve link budgets up to 156 dB (14 dBm – (-141 dBm)) and even up to 168 dB in the United States with a front-end module. This unique link budget allows the LoRa device to break radio communication distance records (more than 1300 km in September 2023).

Robust Against the Doppler Effect

The motion of one radio relative to another will cause a shift in the received frequency due to the Doppler effect. LoRa technology is robust against the Doppler effect and can therefore be deployed in applications with moving assets, such as vehicle tracking, drones, and even satellites. For more information, refer to the application note [AN1200.80: LoRa Modem Doppler Immunity](#).

Robust Against Interference

Due to its chirp spread spectrum nature, LoRa is extremely robust against in-channel or out-of-channel interference, making it particularly suitable for dense or urban deployments. Also, the various SFs are seemingly orthogonal, meaning that they do not interfere with each other even if transmitted at the same RF frequency.

Low Cost

The frequency offset between the transmitter and the receiver reaches +/-25% of the bandwidth without impacting decoding performance. This helps to reduce the price of LoRa transmitters, negating the need for expensive TCXOs, and meaning the crystals embedded in the end-devices need not be manufactured to extreme accuracy.

Different Generations

There are four generations of LoRa chips, with each generation bringing new features and performance improvements.

- Generation 1: SX127x
- Generation 2: SX126x/SX128x and LLCC68
- Generation 3: LR11xx
- Generation 4: LR20xx

LoRa radios can communicate with each other, provided they are properly configured with identical parameters (frequency, spreading factor, bandwidth, and coding rate).

Sensitivity Improvements of the LR2021 Compared to Previous Generations

The radio sensitivity depends on the radio front-end noise figure (NF), the selected BW, and the demodulator signal-to-noise ratio (SNR). This is expressed by the following formula:

$$\text{Sensitivity} = -174\text{dBm/Hz} + \text{NF} + 10\log_{10}(\text{BW}) + \text{SNR (in dB for a given Packet Error Rate (PER))}$$

The SNR improvements between two successive SFs is expressed in the following formula:

$$10 \cdot \log\left(2 \cdot \frac{SF - 1}{SF}\right)$$

The SNR improvements are outlined in the following table:

Table 1. SNR Improvement between two consecutive SFs

SF5 to SF6	SF6 to SF7	SF7 to SF8	SF8 to SF9	SF9 to SF10	SF10 to SF11	SF11 to SF12
2.2 dB	2.3 dB	2.4 dB	2.5 dB	2.5 dB	2.6 dB	2.6 dB

For example, SF6 improves the SNR by 2.22 dB compared to SF5.

The SNR of the LoRa demodulator for a given SF is shown in the following table:

The visible metric is sensitivity, as this directly impacts the communication distance. However, the sensitivity not only integrates the SNR characteristic of the LoRa demodulator, but also the NF of the radio front end.

The following table shows the sensitivity comparison between the generations:

Table 2. Sensitivity Comparison Between Generations

	SX127x	SX126x	LR11xx	LR20xx
SF7 BW 125 kHz	-124 dBm	-124 dBm	-127 dBm	-127.5 dBm
SF12 BW 125 kHz	-137 dBm	-137 dBm	-141 dBm	-141.5 dBm

With the same TX power, the link budget obtained with an LR2021 device is 5 dB better than with an SX127x device. This allows to almost double the communication distance, or to multiply by a factor of 3 the area covered by a LoRa system. It can also double the data rate for the same communication distance, meaning that power consumption is reduced by half.

The LR2021's sensitivity improvements are particularly noteworthy because they are achieved using a direct-tie RF implementation - unlike previous generations of chipsets, which typically required RF switch configurations to optimize receiver sensitivity separately from transmitter performance. The direct-tie architecture in the LR2021 eliminates the need for external RF switching components while simultaneously delivering superior sensitivity performance. This represents a significant improvement for RF designs, where better performance is achieved with reduced complexity and lower BOM cost.

Resistance to Frequency Error: Generation 1, 2, and 3 Versus Generation 4

Frequency offset represents one of the most important design constraints in radio system deployment. When a transmitter and receiver operate at slightly different frequencies, communication reliability deteriorates rapidly. In previous LoRa generations, the frequency offset tolerance was strictly limited to $\pm BW/4$, corresponding to ± 31.25 kHz for BW125. Beyond this frequency offset value, the PER degrades rapidly, as shown in Figure 1:

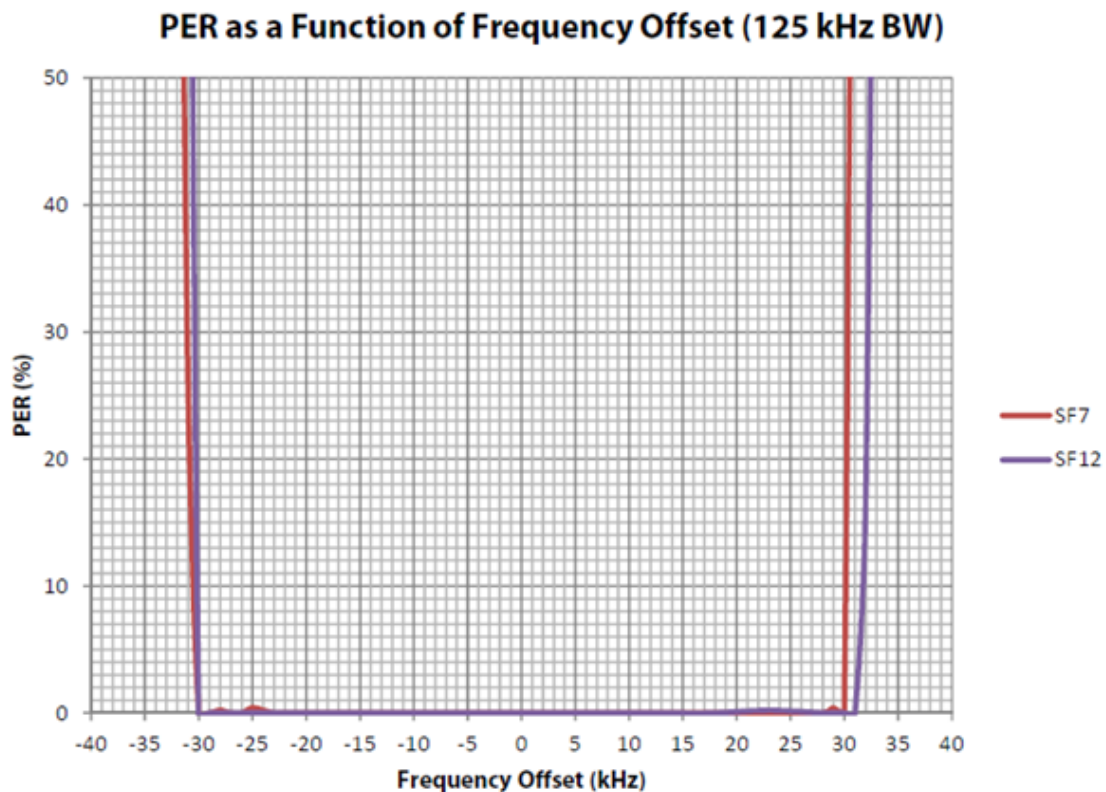


Figure 1. PER as a Function of Frequency Offset

The requirement for high-precision crystal oscillators becomes mandatory for reliable operation over a large temperature range, often necessitating Temperature Compensated Crystal Oscillators (TCXO) on at least one radio in the communication link, generally on the gateway side for LoRaWAN deployments. For LoRa device to device communications, the usage of a TCXO should be avoided as it creates challenges on the power consumption budget, adding 1 to 4 mA of continuous current consumption and increasing the overall solution BOM cost. Careful crystal selection and thermal management considerations allow lower cost or more energy efficient solutions, but still bring system and deployment complexity.

For more information, refer to the following application notes:

- [AN1200.59: Selecting the Optimal Reference Clock](#)
- [AN1200.74: LoRa Edge Clock Requirements](#)

The LR2021 allows for a larger tolerance to frequency offset. This relaxes the constraints on the 32 MHz crystal oscillator, allowing for the use of cheaper crystals, or even crystals on both TX and RX sides.

Generation 4 has two levels of robustness to frequency error:

- Legacy (same as the previous generation): $\pm$ BW/4, for example $\pm$ 31.25 kHz for BW125
- Extended: $\pm$ BW/3, for example $\pm$ 41.7 kHz for BW125

Legacy mode maintains $\pm$ BW/4 tolerance, providing identical performance to previous generations with traditional $\pm$ 31.25 kHz tolerance for 125 kHz bandwidth operations. This mode ensures seamless migration from existing deployments while maintaining backward compatibility.

The new level of frequency offset tolerance, extended mode, provides 33% greater tolerance which relaxes oscillator requirements, enabling selection of lower-cost crystals, a key element in IoT deployments where device economics are primary decision factors.

Both levels of frequency offset tolerance significantly improve LoRa system design flexibility.

The configuration of the level of robustness to frequency error is made via a register change, writing in bits 16 and 17 of register 0x00F30A2C, for example via the pseudo code command:

```
// Configure mode at address 0xF30A2C, bits [17:16]

WriteRegMemMask32(addr = 0xF30A2C, Mask = (3 << 16), Value = (mode << 16));

// where mode = 0 (legacy) or 1 (extended)
```

Resistance to Doppler Shift and Doppler Rate

The Doppler effect presents a fundamental challenge in wireless communications when transmitters and receivers experience relative motion. This phenomenon manifests in two distinct forms that impact LoRa communication links. Doppler shift represents the instantaneous change in received frequency due to relative speed between communicating devices, while Doppler rate refers to the rate of change of this frequency shift over time, typically caused by acceleration or changing velocity vectors.

In traditional wireless systems, Doppler effects can cause complete communication failure when the frequency shift exceeds the receiver's tracking capability. LoRa's unique chirp spread spectrum modulation provides inherent robustness against these effects, but the degree of tolerance has evolved significantly across generations, with each improvement enabling new categories of mobile applications. For more information, refer to the application note [AN1200.80: LoRa Modem Doppler Immunity](#).

Generation 1 and 2 receivers evaluate Doppler over the entire packet duration, depending on the SF and BW. These receivers typically support spreading factors up to SF8 in challenging Doppler environments.

Generation 3 improves the frequency drift tolerance compared to Generation 1 and Generation 2, as Doppler is evaluated over individual symbol periods. The symbol-level approach provides highly improved Doppler tolerance by reducing the evaluation window from packet duration (potentially seconds) to symbol duration (typically milliseconds). This provides a much greater robustness to Doppler effects, allowing a robust communication up to SF10 or SF11, even in demanding scenarios like satellite communications.

Compared to Generation 3, Generation 4 improves the tracking range, which is the ability to follow and adjust to frequency changes over time, when the LDRO feature is enabled. Generation 4 also has improved frequency drift tolerance when LDRO is enabled, built upon the symbol-level evaluation approach established in Generation 3, but providing better frequency tracking capability during extended symbol periods. When LDRO is enabled, the modem can maintain frequency lock more effectively throughout long symbol durations, improving communication reliability in dynamic frequency environments. This leads to significant frequency drift tolerance for SF11/12, where LDRO is typically enabled.

Increased Resistance to Bursty Interference

Interference Mechanisms and LoRa Immunity

In dense urban deployments and industrial environments, multiple radio systems operating simultaneously create challenging interference scenarios for wireless communications. The interference manifests in two primary forms: in-channel (co-channel) interference occurring on the same frequency as the intended signal, and out-of-channel (adjacent channel) interference from nearby frequency bands. LoRa's inherent robustness against both types of interference results from its Chirp Spread Spectrum (CSS) modulation approach.

First, information bits are distributed across the entire LoRa bandwidth rather than concentrated at specific frequencies, providing spectral spreading. Moreover, data is encoded over long symbol durations, allowing reconstruction even when portions of the signal are corrupted.

This makes LoRa particularly effective against narrowband bursty interference, as the interference energy affects only a fraction of the total signal energy spread across frequency and time domains.

Coding Rate Enhancement

LoRa employs Forward Error Correction (FEC) through coding rates that add redundancy for error recovery. The coding rate expresses the ratio of useful information bits to total transmitted bits:

- CR 4/5: 4 information bits per 5 transmitted bits (20% redundancy)
- CR 4/6: 4 information bits per 6 transmitted bits (33% redundancy)
- CR 4/7: 4 information bits per 7 transmitted bits (43% redundancy)
- CR 4/8: 4 information bits per 8 transmitted bits (50% redundancy)

Therefore, higher coding rates provide increased error correction capability at the expense of longer transmission times and consequently higher power consumption.

Introduced in Generation 3 (LR11xx) and enhanced in Generation 4 (LR2021), long interleaving distributes coded bits across the entire LoRa frame duration. This technique allows frame reconstruction even when consecutive LoRa symbols are corrupted, providing superior resilience against frequency-selective fading, burst interference longer than single symbol duration, and also fast fading.

The LR2021 adds Long Interleaver Convolutional code 4/6 and code 4/8 (called CR8 and CR9 later in this document), which represents a significant breakthrough in LoRa interference mitigation. Convolutional coding spreads each data point across multiple adjacent coded symbols, creating overlapping protection that maintains signal integrity even under severe interference conditions.

CR8 is a 2/3 coding rate with both long interleaver and convolutional encoding. It transmits 3 coded bits for every 2 information bits, creating 33% redundancy overhead. Each information bit is protected across 3-4 symbols.

CR9 is a 1/2 coding rate with both long interleaver and convolutional encoding. It transmits 2 coded bits for every 1 information bit, thus 100% redundancy overhead. It provides the maximum protection against interference since each information bit is strongly protected across 5-6 symbols.

The legacy coding rates, the Long Interleaver, and the Convolution coding rates are configurable in the command `SetLoRaModulationParams(...)`.

Resistance to Fast Fading

Fast fading occurs when the radio signal strength changes rapidly due to the relative motion between transmitter and receiver, or changes in the propagation environment. Fast fading physically results from multiple environmental effects, always present in real world deployment, with moving obstacles, rotating machinery, and so on: multipath propagation (radio signals taking multiple paths to reach the receiver), Doppler effect (frequency shifts due to relative motion), and constructive/destructive interference.

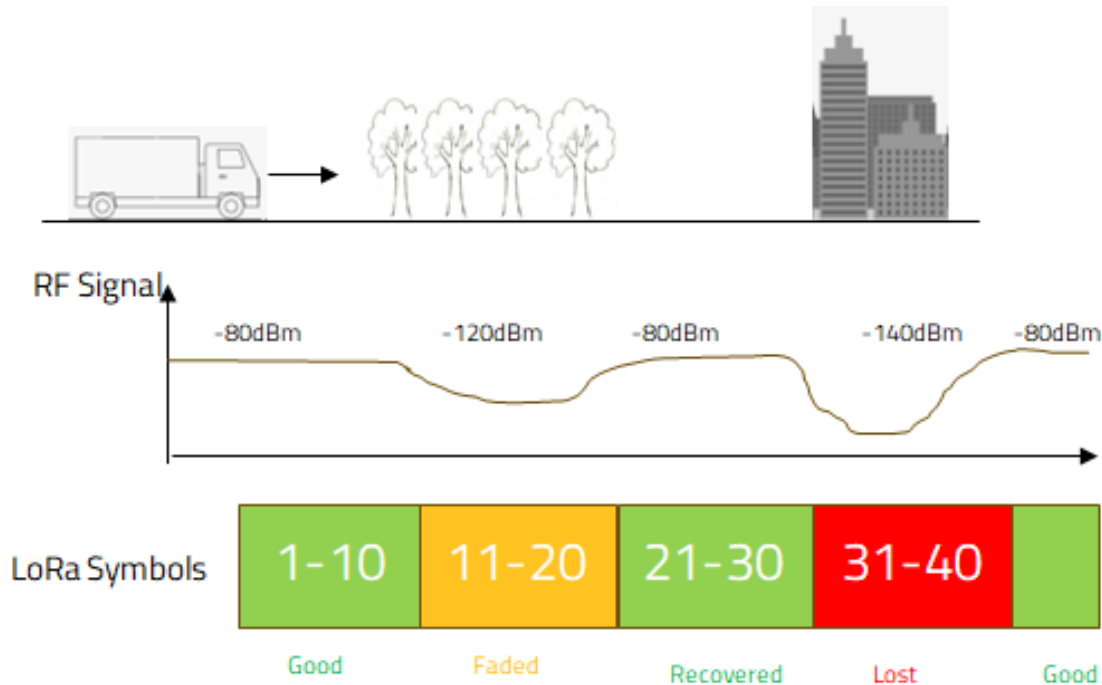


Figure 2. Fast Fading

The long interleaver feature in the LR11xx series provides robustness against fast fading, as the information is spread out over the LoRa frame. However, losing a series of successive LoRa symbols in an incoming LoRa frame compromises its good reception, even with long interleaver enabled. The LR2021's new coding schemes, convolutional coding CR8 (2/3) and CR9 (1/2), are more robust against this effect, since the information contained in each symbol is spread over a window of successive symbols.

Convolutional coding can be affected by long error bursts, but when combined with a long interleaving scheme, the errors are distributed across the entire packet rather than concentrated in one area, allowing the convolutional coding to effectively correct them. Used together, the long interleaver and convolutional code provide extreme robustness against fast fading or general interference effects.

The new coding rates CR8 and CR9 activate both the long interleaver and the convolutional coding features. Similar to the legacy coding rates CR4/5 to 4/8, they add redundancy in the LoRa frame (2/3 for CR8 and 1/2 for CR9) and therefore increase the time on air.

Blocking Performance

RF interference can occur both within the selected channel (in-band interference) and outside of it (out-of-band interference). When interference occurs outside the selected channel, LoRa's inherent robustness allows it to support very strong interfering signals without degradation of the LoRa radio link performance. These very strong out-of-channel interferers are referred to as blockers or blocking signals. The Blocking performance test consists in applying a continuous unmodulated interfering signal at various frequency offsets from the wanted LoRa signal, with the blocker power progressively increased until the LoRa Packet Error Rate reaches 30%. The test is conducted with the wanted signal set at sensitivity + 3 dB to ensure realistic operating conditions. This methodology characterizes the receiver's selectivity - its ability to discriminate between the desired signal and unwanted interference at different frequency separations.

The following figure compares the blocking performance of the LR2021 (solid lines) and LR11xx (dotted lines) over a frequency offset range of -2 MHz to +2 MHz, using a continuous unmodulated blocking signal for SF7 BW125 at 915 MHz.

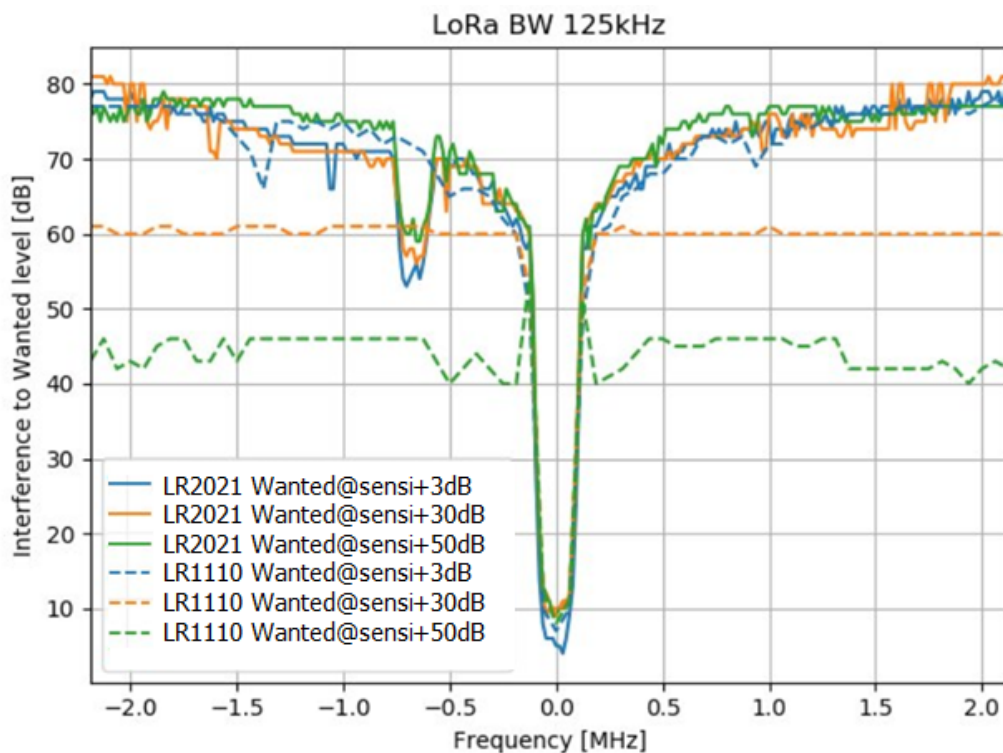


Figure 3. Blocking Performance of LR2021 SF7 BW125 at 915MHz

The figure shows that the LR2021 can support blocking signals 70 dB stronger than the wanted LoRa signal at 1 MHz offset, and 80 dB stronger at 2 MHz offset, achieving a similar performance to the LR1110 for small wanted signals, however the LR2021 maintains this performance even if the wanted signal level is increased, which means that the LR2021 demonstrates a superior blocking performance when close to saturation.

Note that the blocking performance is determined by the cumulative performance of the entire reception chain, from the RF front-end components (Low Noise Amplifier, anti-aliasing filter, mixer, Automatic Gain Control) through to the digital signal processing blocks and LoRa demodulator. The LR2021 incorporates multiple improvements throughout the receive chain which complement each other to deliver superior blocking immunity compared to previous generations of LoRa devices, especially in the presence of strong wanted and unwanted signals. Combined with LoRa's inherent spread-spectrum advantages, the LR2021 features exceptional interference rejection capabilities that significantly exceed those of conventional narrowband RF receivers.

Multi-SF Detection

The LR2021's fourth-generation LoRa IP introduces multi-SF detection capability, fundamentally changing how LoRa devices can handle mixed spreading factor environments. Multi-SF CAD can detect LoRa symbols on multiple SFs in parallel. If RX is configured as CadExitMode, it automatically switches to RX mode to demodulate the detected SF. The system uses a main LoRa detector plus up to three side detectors, each operating independently but sharing the same bandwidth and RF channel.

Multi-SF detection operates under specific technical constraints to ensure optimal performance:

- Spreading Factor Constraints
 - Up to 4 SFs can be detected simultaneously (main SF + 3 side SFs)
 - The main SF (configured with SetLoraModulationParams) must be the smallest SF in the detection, or the largest for multi-SF CAD operation
 - All configured SFs must be different; duplicate SFs are not permitted
 - The difference between MaxSF and MinSF must be ≤ 4
- Bandwidth-Dependent Limitations
 - BW > 500 kHz:
 - Only 2 side SFs allowed (3 SFs in total)
 - If main SF is 10, 11, or 12 with BW > 500 kHz, only 1 side SF is permitted (2 SFs in total)



After a successful reception, it is possible to know which SF was demodulated with the command GetLoraPacketStatus. Calling SetLoraModulationParams disables all side SFs, and the user will have to re-configure them.



The main SF is configured using command SetLoraModulationParams.



The side SF is configured using command SetLoraSideDetConfig.

Channel Activity Detection (CAD)

Wireless applications often need to search for air radio traffic, either to detect the wanted signal (wake-on-radio), or to avoid collision (as for Listen Before Talk (LBT) or Carrier Sense Multiple Access (CSMA)). As LoRa allows the reception of signals below the noise floor, RSSI-based measurements alone will not detect all LoRa signals. The Channel Activity Detect (CAD) function can be used instead. CAD detects any LoRa communication packet in the air, for a given SF and bandwidth setting, including traffic below the noise floor to within approximately 3 dB of the sensitivity of the receiver. The CAD process is optimized to perform LoRa packet detection quickly, using as little energy as possible. Depending on the configuration, it can be automatically followed by a TX (for LBT/CSMA) or by a RX (for wake-on-radio).

The LR2021 CAD implementation has the following improvements compared to CAD on the previous generation of devices:

- CAD can be configured to detect LoRa preamble symbols and/or LoRa payload symbols
- It performs a CAD on multiple SFs in parallel (multi-SF)
- Fast CAD provides optimization of power consumption when no LoRa signal is present

Those improvements are detailed in the following sections.



The LR2021 can perform a CAD on GFSK frames. This functionality supplements basic RSSI detection. Details of this feature are not addressed in this document.

Preamble Only CAD

The SX127x devices can perform CAD on the preamble symbols. All other LoRa products (SX126x, SX128x, LR11xx) can detect any symbol in the entire packet, regardless of whether the symbols are unmodulated (preamble) or modulated (payload).

The LR2021 CAD system provides selectable detection modes through the `pbl_any` parameter in the `SetLoraCadParams` command. When configured for preamble-only detection (`pbl_any=1`), the system focuses exclusively on unmodulated LoRa preamble symbols, enabling faster transition to reception mode upon successful detection. This approach leverages the deterministic nature of preamble chirps to achieve reduced detection latency compared to full-packet symbol detection.

Table 3. Preamble CAD vs Symbol CAD Comparison

Parameter	Preamble-only CAD (<code>pbl_any=1</code>)	Full-Packet Symbol CAD (<code>pbl_any=0</code>)
Detection Target	Unmodulated preamble symbols only	All LoRa symbols (preamble + payload)
Symbol Characteristics	Deterministic chirp pattern	Variable modulated data patterns
Ongoing Transmission Detection	Limited - only if preamble not missed	Complete- can detect midpacket
Detection Window	Packet start only	Entire packet duration
Optimal Use Cases	Wake-on-radio, CAD to Rx	Network monitoring, spectrum analysis
False Detection Probability	Lower in noise conditions	Higher potential at low SF/high SNR
Synchronization Advantage	Immediate packet duration knowledge	Requires additional duration search

Preamble Only vs Symbol CAD Comparison

Preamble-only detection enables immediate transition to sync word search upon successful preamble detection, without requiring analysis of the variable packet data that follows. Both preamble and modulated symbol detection achieve lower power consumption compared to full RX mode by completing the detection process in just a few symbols rather than receiving entire packets. These advantages are particularly beneficial in battery-powered applications where minimizing the energy consumption during CAD directly impacts operational lifetime.

In contrast, CAD on LoRa symbols allows to detect a LoRa packet from transmissions already in progress. This also offers improved detection probability in high signal-to-noise ratio conditions due to higher correlation energy available from modulated payload symbols.

Multi-SF CAD

Multi-SF CAD extends the parallel detection capabilities of the LR2021's multi-SF to CAD operations, enabling simultaneous monitoring across multiple spreading factors within a single CAD cycle. The system employs a similar detection architecture as multi-SF reception, utilizing one main detector plus up to three side detectors, each configured for independent spreading factor operation while sharing common bandwidth and channel parameters.

Multi-SF CAD configuration follows the standard multi-SF setup sequence with CAD-specific parameter optimization:

- `SetPktType()` - Configure packet type to LoRa mode
- `SetLoraModulationParams()` - Set main detector parameters (SF, bandwidth, coding rate)
- `SetLoraPacketParams()` - Configure packet parameters and header settings
- `SetLoraCadParams()` - Define CAD-specific behavior (detection duration, symbol targeting, fast-CAD parameters)
- `SetLoraSideDetConfig()` - Enable and configure up to three side detectors for additional spreading factors
- `SetLoraCad()` - Initiate the configured multi-SF CAD operation

Fast CAD

CAD aims to detect a LoRa signal out of noise as quickly as possible, meaning on as few LoRa symbols as possible. This is done by comparing the received signal with a given peak-to-noise ratio threshold: If the signal detected is above the threshold, then the signal is likely to be a real LoRa signal. A threshold that is too high might discard weak signals, resulting in detection failure. On the contrary, a threshold that is too low might wrongly detect a LoRa signal out of noise, resulting in an incorrect detection.

The following figure shows how the fast CAD works to detect 5 LoRa symbols. The LR2021 fast CAD feature progressively increases the peak-to-noise threshold for CAD detection for the successive symbols to be used for the LoRa signal detection. If an actual LoRa packet is detected, the whole process is executed until the end. This allows the CAD process to stop quickly if no LoRa packet is detected, instead of trying to detect 5 successive symbols.

On average, the fast CAD process stops after 1.5 symbols if there is no LoRa signal present. Fast CAD reduces the power consumption of the CAD process, allowing a more power-efficient LBT/CSMA process.

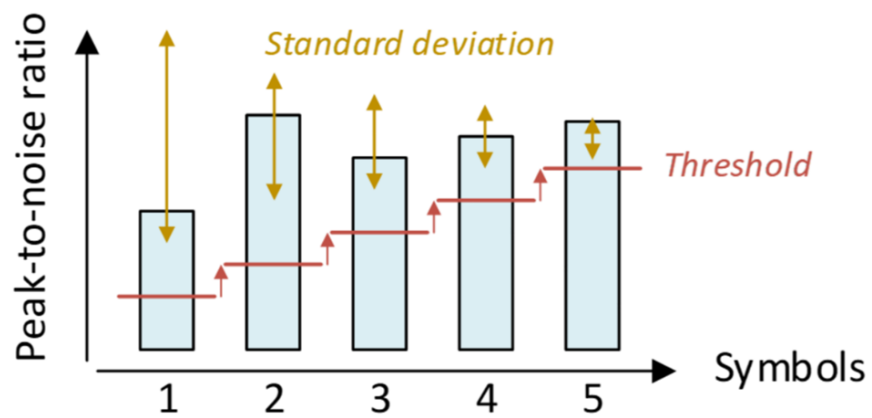


Figure 4. Fast CAD Threshold Adjustment Principle

Summary

Compared to previous generations, the LR2021 device has the following benefits:

- Improved radio sensitivity
 - With the same TX power, the link budget obtained with LR2021 is 5 dB better than with an SX127x device.
- Larger tolerance to frequency offset
 - This relaxes the constraints on the 32 MHz crystal oscillator, allowing for the use of cheaper crystals.
- Improved resistance to Doppler change and Doppler shift
 - LR2021 improves the double tracking range when LDRO is enabled. This leads to significant frequency drift tolerance for SF11/12.
- Increased resistance to bursty interference
 - LR2021 adds convolutional code (CR8 and CR9), further improving the robustness to interferers.
- Improved blocking performance
 - LR2021 support blocking signals 70 dB stronger than the wanted LoRa signal at 1 MHz offset and 80 dB stronger at 2 MHz offset.
- Improved phase noise performance
 - LR2021 complies with ARIB without compromising the TX power.
- Extreme robustness against fast fading
 - LR2021 has a new, more resistant coding scheme called convolutional coding: CR8 (2/3) and CR9 (1/2).
- Improved CAD implementation
 - The LR2021 CAD function can be configured to detect LoRa preamble symbols or LoRa payload symbols.
 - Multi-SF CAD can listen to LoRa symbols on multiple SFs in parallel.
 - Fast CAD provides optimization of power consumption for false detections.

LoRa Performance Summary Across Generations

Table 4. LoRa Performance Summary Across Generations

Feature	SX1272 (Gen 1)	SX1261/2 (Gen 2)	LR1110 (Gen 3)	LR2021 (Gen 4)
LoRa Sensitivity SF7	-124 dBm	-124 dBm	-127 dBm	-127.5 dBm
LoRa Sensitivity SF12	-137 dBm	-137 dBm	-141 dBm	-141.5 dBm
Max Data Rate	~62.5 kbps	~62.5 kbps	~62.5 kbps	125 kbps
Frequency Offset Tolerance	$\pm BW/4$	$\pm BW/4$	$\pm BW/4$	$\pm BW/3$ (Extended mode)
CAD Capabilities	Preamble only	Standard	Standard	Multi-SF + Fast CAD
Doppler Tolerance	Packet-level	Packet-level	Symbol-level	Enhanced Symbol + LDRO
Coding Rates	CR 4/5 to 4/8	CR 4/5 to 4/8	CR 4/5 to 4/8 + Long	CR 4/5 to 4/8 + Long + Convolutional

Blocking Immunity	Standard	Standard	Enhanced	70-80 dB @ 1-2 MHz offset
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Compatibility Matrix: SF5-6 with Generation 1, 2 and 3

LoRa radios can communicate with each other, provided they are properly configured with identical parameters (frequency, spreading factor, bandwidth, coding rate, LoRa Sync Word, and chirp polarity). This allows for the adoption of new generations without being forced to rewrite the application code or redesign the air protocol.

The following table outlines LoRa radio compatibility details:

Table 5. LoRa Radio Compatibility Details

Radio Generation	Gen 1	Gen 2	Gen 3 and 4
Compatibility	- Supports SF6 with implicit header and special interleaving scheme - Does not support SF5	- Supports SF5 and SF6 with implicit and explicit header and standard interleaving scheme - Communication between Gen1 and Gen2 at SF6 is not possible	- Supports SF5 and SF6 (including legacy SF6 of Gen1) - Supports both formats with a static configuration bit to communicate with Gen1 at SF6 implicit header ^[1] - Compatibility mode with syncword format used in Gen1 (Gen 4 only)

[1] Please refer to the LR2021 driver on Github/LoRa-Net for an implementation example

Glossary

Term	Description
BW	Bandwidth
CAD	Channel Activity Detect
CR	Coding Rate
CSMA	Carrier Sense Multiple Access
FEC	Forward Error Correction
GFSK	Gaussian Frequency Shift Keying
HVAC	Heating, Ventilation, and Air Conditioning
IoT	Internet Of Things
LBT	Listen Before Talk
LDRO	Low Data Rate Operatio
LEO	Low Earth Orbit
LoRa®	Long Range Communication. The LoRa® Mark is a registered trademark of the Semtech Corporation
LoRaWAN®	LoRa Wide Area Network
NF	Noise Figure
PHY	Physical Layer
PER	Packet Error Rate
RF	Radio Frequency
Rx	Reception
SF	Spreading Factor
SNR	Signal to Noise Ratio
Tx	Transmission

Revision History

Version	ECO	Date	Changes and/or Modifications
1.0	076680	October 2025	Preliminary Version

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